

Revised Generation Interconnection System Impact Study Report

For Queue Project Q19-1316

Bear Island 138 kV
June 20, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1316. The Interconnection Customer proposes a 75.5 MW solar photovoltaic facility interconnecting at the Bear Island 138 kV substation in the Virginia area. The request was received on October 22, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 384 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$298,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of Leavitt-Birchwood 345.0 kV, the Fluvanna P.S-Bear 345.0 kV loads to 101.3% of its 1,358 MVA rating. Project DFAX: 8.9%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Fluvanna P.S-Bear 345.0 kV	Reconductor line	\$67,604,000	0% (\$298,000)
	Total allocated to project		\$298,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.